MARS MISSION PROJECT

Earth–Mars Beacon Relay Concept

Distance, travel time, communication beacons, current NASA practice, and a 10x analysis



Riad · October 2026

THE CONCEPT IN NUMBERS

Executive summary

218M km

Earth–Mars distance today
(28 Oct 2026) · 136M miles

54.6M km

Nearest possible distance
(record: 55.8M km, Aug 2003)

2,261 h

Trip today at a constant
60,000 mph (~94 days)

1,131

Beacons today, one every 2 h
(120,000 mi apart)

1,878

Beacons for worst case
(Mars at 225M mi)

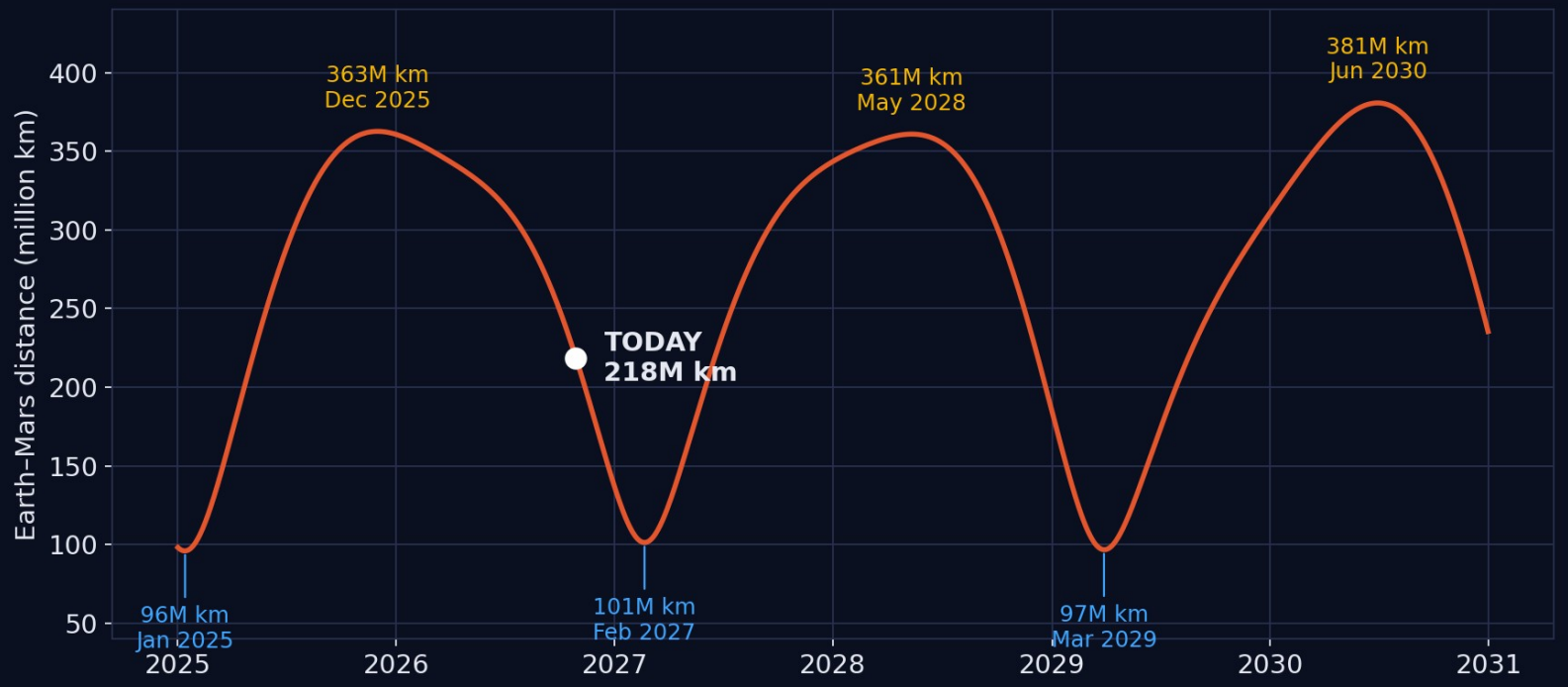
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Beacons NASA uses today:
direct links + ~5 Mars orbiters

- **Feasible on paper:** the beacon count is simple arithmetic: miles ÷ 120,000.
- **Hard in practice:** beacons drift apart in solar orbit, the Sun blocks the line every ~26 months, and ~1,900 spacecraft is far beyond any program to date.
- **Recommended path:** a hybrid of 3–10 high-power relays, with the chain idea kept for near-Earth/near-Mars segments.

DISTANCE ANALYSIS · NASA JPL DE432 EPHEMERIS

How far is Mars? It depends on the date

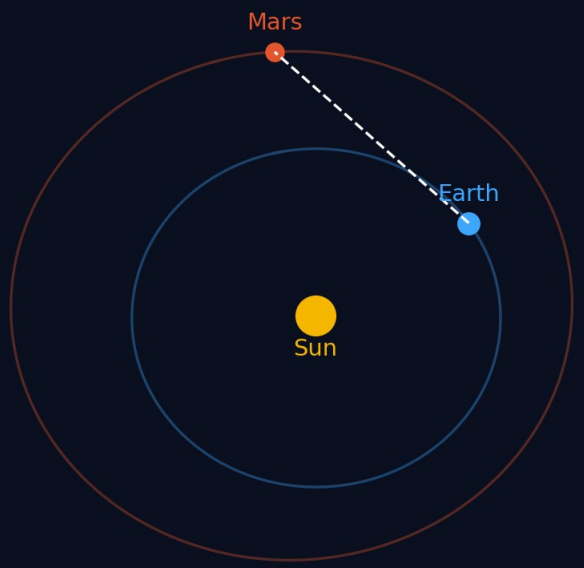


- **Cycle:** closest approach every ~26 months (synodic period).
- **Range:** from ~55M km (best case) to ~400M km (Mars behind the Sun).
- **Today:** 218M km, closing ~38M km per month.
- **Next closest:** ~20 Feb 2027 at 101M km, a "distant" close approach because Mars's orbit is oval.
- **Signal delay:** 5–21 min one way (12.1 min today).

ORBITAL GEOMETRY

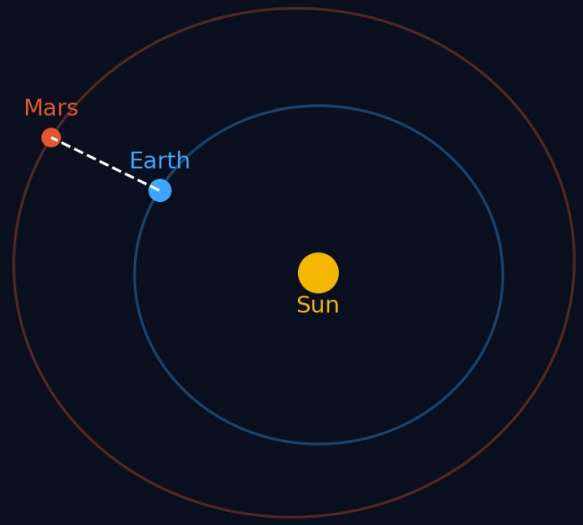
Where the planets are

Positions today
28 Oct 2026 · 218 million km



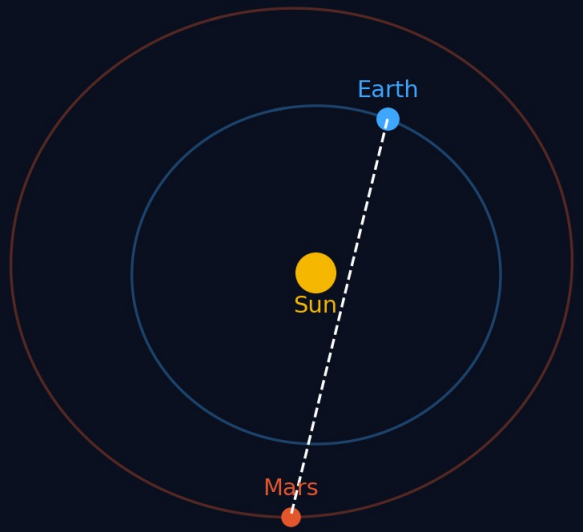
Today: Earth is catching up with Mars

Next closest approach
20 Feb 2027 · 101 million km



Feb 2027: same side of the Sun = closest

Solar conjunction (Sun in the way)
30 Nov 2025 · 363 million km



Dec 2025: Sun in between = farthest + blackout

Key insight

Earth orbits faster (1 yr) than Mars (1.88 yr), so the gap keeps changing. Any fixed "chain" between the two planets has to stretch, bend and re-aim continuously.

CLOSE-APPROACH ANALYSIS

Nearest distances between Earth and Mars

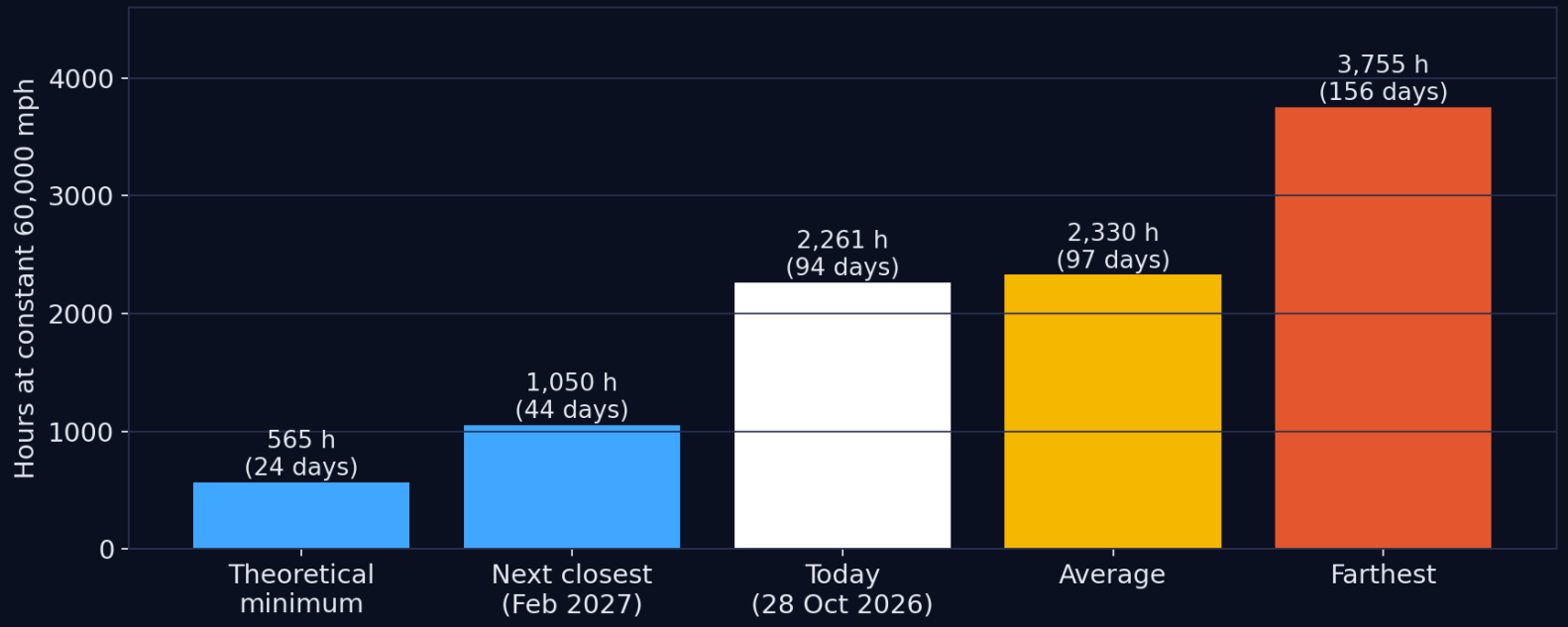
Event	Date	Distance (km)	Distance (miles)	Hours at 60,000 mph	Beacons (1 per 2 h)
Theoretical minimum	—	54.6 M	33.9 M	~565 h (24 d)	~283
Closest in ~60,000 yrs	27 Aug 2003	55.8 M	34.6 M	~577 h	~289
Recent close pass	31 Jul 2018	57.6 M	35.8 M	~597 h	~298
Next close approach	**~20 Feb 2027**	**101.4 M**	**63.0 M**	**~1,050 h (44 d)**	**~525**
Next very close pass	Sep 2035	~56.9 M	~35.4 M	~589 h	~295
Record next beaten	28 Aug 2287	< 55.8 M	< 34.6 M	—	—

Takeaway

Launching around a very close pass (like 2035) roughly halves the trip and the beacon count versus an average pass.

TRAVEL ANALYSIS

Travel time at a constant 60,000 mph



Formula

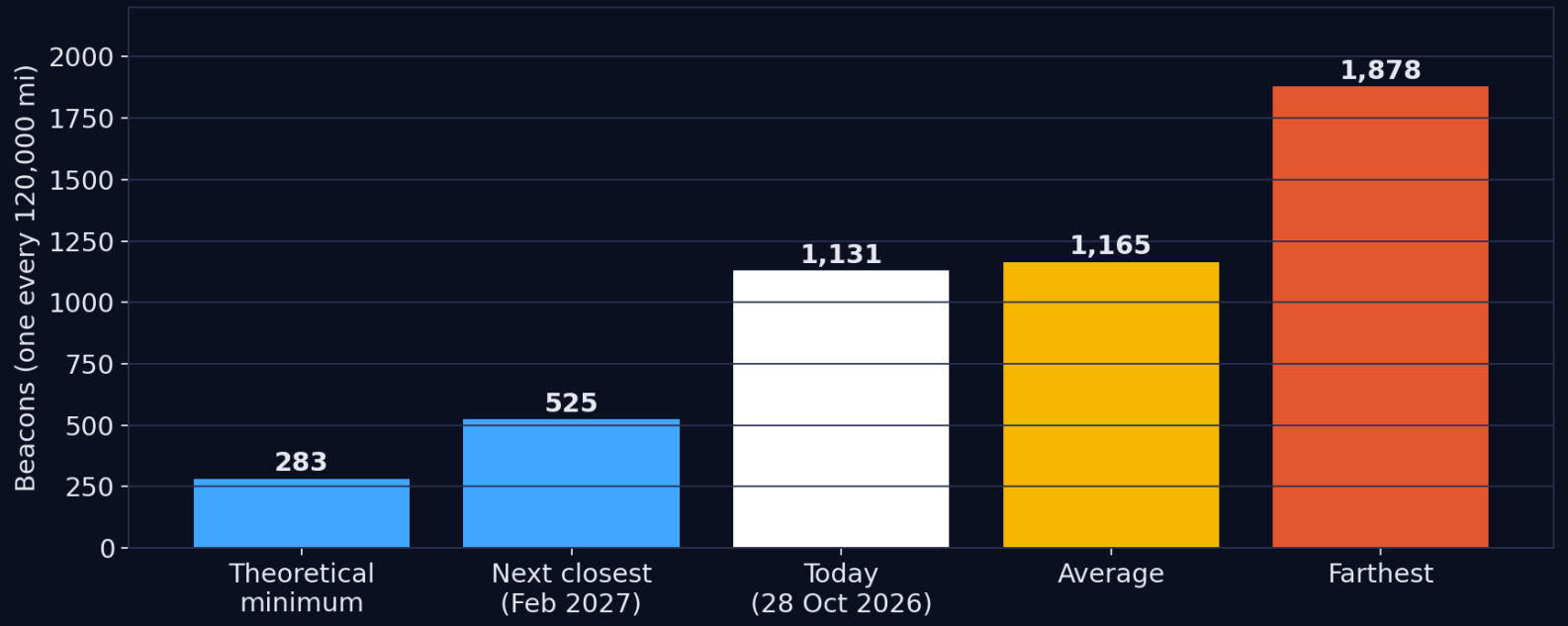
hours = miles ÷ 60,000

Today: 135,668,857 ÷ 60,000 ≈ 2,261 h

- **Speed check:** 60,000 mph ≈ 26.8 km/s, similar to Earth's own orbital speed around the Sun (~67,000 mph).
- **Reality:** Mars moves during the trip, so ships aim ahead and coast on a curved path. Real trips take 7–9 months.
- **Best value:** depart so you arrive near a close approach.

COMMUNICATIONS ANALYSIS

Beacon chain sizing: one beacon every 2 hours



Formula

spacing = 60,000 mph × 2 h = 120,000 mi

beacons = miles ÷ 120,000

- **Design number:** ~1,900 beacons (1,878 for worst case + spares).
- **Per-hop distance:** 120,000 mi ≈ half the Earth–Moon distance.
- **Coverage rule:** each beacon should reach 2 hops (240,000 mi) so one failure doesn't break the chain.

RISK ANALYSIS

Engineering challenges

Challenge	Why it matters	Mitigation
Orbital drift	Each beacon orbits the Sun at its own speed, so the chain spreads and bends within weeks	Ion thrusters for station-keeping, or fewer relays in stable orbits
Solar conjunction	Every ~26 months the Sun sits between Earth and Mars: ~2-week blackout	Off-axis relays (e.g. Sun–Earth L4/L5 or leading/trailing Mars orbit)
Power	Sunlight at Mars is only ~43% of Earth's ($1 \div 1.52^2$)	Larger solar arrays toward the Mars end
Signal delay	Beacons do NOT make radio faster: still 5–21 min (speed of light)	Autonomous rovers/landers; store-and-forward networking
Chain failure	A single dead beacon can cut the link	2-hop range, spares, mesh routing
Scale & cost	~1,900 spacecraft to launch and operate	Mass-produced CubeSat-class units, rideshare launches

LINK ANALYSIS

Why beacons help: signal strength per hop

~1,130×

shorter per hop: 120,000 mi vs 135.7M mi direct

~1.3 million×

stronger signal per hop (distance²), ≈ 61 dB

0 min

delay saved: light still needs 12.1 min today

- **Small radios suffice:** with 61 dB less path loss, each beacon could use a small antenna and a few watts instead of a 70 m dish.
- **Higher data rates:** shorter hops allow much faster links than today's few-Mbit/s direct links.
- **Trade-off:** the gain is offset by ~1,900 points of failure, plus station-keeping and power for every unit.
- **Note:** signal strength falls with distance squared (free-space path loss). Figures are idealised and ignore antennas, noise and pointing.

CURRENT PRACTICE

What NASA uses today: zero beacons

Element	How many	Role
Deep Space Network (Earth)	3 sites: California, Spain, Australia · about a dozen 34 m / 70 m dishes	One site always faces Mars as Earth rotates
Mars relay orbiters	~3 NASA (MRO, Odyssey, MAVEN) + 2 ESA (Mars Express, TGO)	Collect rover data and forward it to Earth
Beacons in between	0	—
Proposed upgrade	1 Mars Telecommunications Orbiter	One powerful satellite to boost data rates

Why it works without relays

Huge, sensitive dishes on Earth can hear weak signals across 400M km. Voyager 1 is still heard from 24+ billion km with no relays at all.

MARS ORBIT CENSUS

Satellites currently orbiting Mars

Spacecraft	Agency	Arrived	Relays rover data?
Mars Odyssey	NASA (USA)	2001	Yes · oldest, low on fuel
Mars Express	ESA (Europe)	2003	Yes
Mars Reconnaissance Orbiter	NASA	2006	Yes · main relay
MAVEN	NASA	2014	Yes · contact lost late 2025 (reported)
Trace Gas Orbiter	ESA / Russia	2016	Yes · main relay
Hope	UAE	2021	No · science only
Tianwen-1 orbiter	China	2021	Yes · for Zhurong rover

- **~7–8 active** spacecraft from 5 agencies.
- **~8–10 dead** spacecraft still circling (Mariner 9, Vikings, MGS, Mangalyaan, Soviet probes).
- **Incoming:** NASA ESCAPADE (2 small sats), arriving ~2027.
- **Natural moons:** Phobos and Deimos.
- Status is from general knowledge; aging orbiters change often.

OPTIONS ANALYSIS

Concept comparison

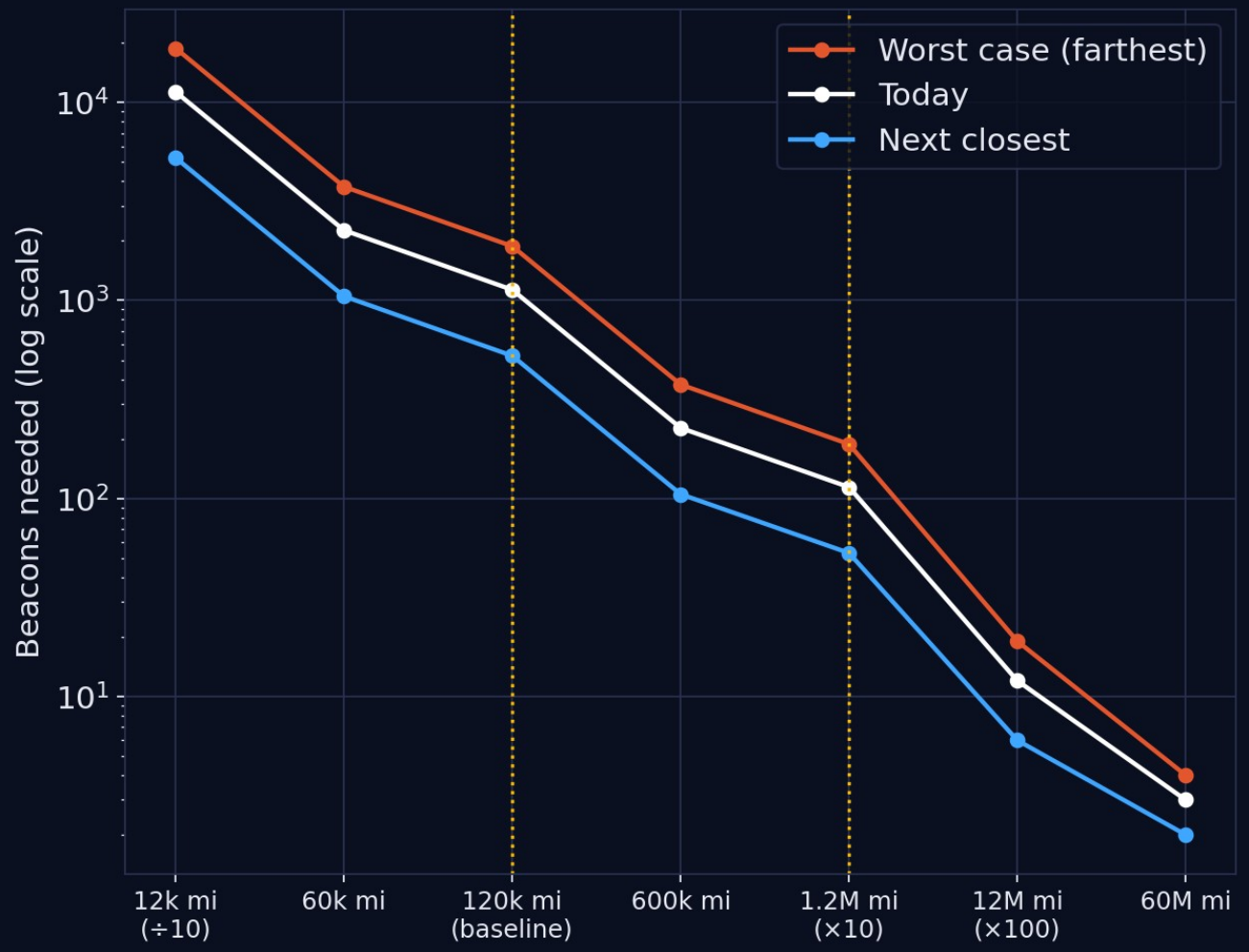
Criterion	Beacon chain (~1,900)	Few relays (3–10)	Direct link (today)
Spacecraft needed	~1,900	3–10	0 extra
Data rate potential	Very high	High	Low–medium
Conjunction coverage	Only with off-axis branch	Yes, with off-axis relay	No (~2-week blackout)
Signal delay	5–21 min (unchanged)	5–21 min (unchanged)	5–21 min
Complexity	Very high	Medium	Low
Relative cost (qualitative)	Very high	Medium	Lowest (exists)
Resilience	Many small failure points	Few critical units	Depends on orbiters

Verdict

The few-relays option captures most of the data-rate benefit at a fraction of the scale. That is why real mission studies favour it.

SENSITIVITY ANALYSIS

10x analysis: what if we scale by ten?



Scenario	Trip today	Beacons today	Worst case
Baseline: 60k mph, every 2 h	2,261 h	1,131	1,878
10× speed: 600k mph, every 2 h	226 h (9.4 d)	114	188
10× spacing: every 20 h	**2,261 h**	**114**	**188**
÷10 spacing: every 12 min	2,261 h	11,306	18,775

- **Beacon count scales 1:1** with spacing: 10× farther apart = 10× fewer beacons.
- **10× speed** (600k mph) beats the fastest craft ever (Parker Solar Probe, ~430k mph) and isn't realistic for Mars yet.
- **Sweet spot:** 10× spacing (1.2M mi) is still ~41 dB better than a direct link, with only ~190 beacons.

PLAN

Recommendations & next steps

- **Adopt a hybrid design:** 2–3 relays near Earth, 2–3 in Mars orbit, plus 1–2 off-axis relays to bridge solar conjunction.
- **If keeping the chain:** use 10× spacing (~190 beacons, 1.2M mi apart) instead of ~1,900.
- **Target a good window:** aim for a close approach (e.g. 2029 or 2035) to cut trip time and link distance.
- **Size the link budget:** choose a target data rate, then antenna size and power per relay.
- **Build a cost model:** per-unit cost × count + launch + operations, compared across the 3 options.
- **Interactive model:** animate Earth, Mars and the relay chain to visualise drift and conjunction.

~1,900

beacons: original chain (1 every 2 h)

~190

beacons: chain at 10× spacing (1 every 20 h)

~5–8

relays: recommended hybrid design

REFERENCE

Sources & assumptions

- **Distances:** computed with NASA JPL DE432 ephemeris (astropy) for 2025–2031. Today = 28 Oct 2026, 218.3M km.
- **Historic/limit values:** published figures (2003 record 55.76M km; theoretical minimum ~54.6M km; 2287 next record).
- **Travel model:** constant 60,000 mph, straight line, Mars treated as stationary. Simplified, not a real trajectory.
- **Beacons:** one per 2 h of travel = 120,000 mi spacing; counts rounded up.
- **Link gain:** free-space path loss ∝ distance^2; antenna, noise and pointing ignored.
- **Spacecraft status:** general knowledge, not a live check; orbiter status may have changed.
- **Cost and complexity:** qualitative comparisons only, not estimates.